

Pultruded FRP for Offshore, Floating & Saline PV

Saltwater, brackish fishery-PV ponds and saline-alkali tidal flats corrode galvanised steel and aluminium racking from day one — driving sacrificial anodes, cathodic protection, hot-dip coatings and endless repaint. F1 Composite pultruded FRP posts, rails and decking are corrosion-immune in chloride and alkaline service, non-conductive and UV-stable — removing the anode, coating and grounding scope for fixed-offshore, floating and saline ground-mount arrays.



Fishery-PV (aquavoltaic) racking in FRP over brackish ponds



Coastal / offshore array — no anode, no galvanizing, no repaint

CORE PERFORMANCE REQUIREMENTS

Chloride / Salt Immunity

Corrosion-immune in seawater & salt spray; no anode

Alkaline Soil Service

Resists saline-alkali tidal-flat & sabkha soils

No Cathodic Protection

No sacrificial anode or impressed-current system

Non-Conductive

No grounding / bonding; ends galvanic corrosion

Floating-PV Ready

Stable on pontoons; UV & humidity resistant

Low-Maintenance OPEX

No repaint / re-galvanize; 25+ yr in marine air

STANDARDS & CODES

EN 13706 Structural pultruded profiles (E23)

ASTM D3917 Pultruded profile dimensional tolerance

ASTM D790 Flexural strength & modulus

ASTM D543 Chemical / corrosion resistance

ASCE 7 Wind & snow structural load basis

DNV-ST-C501 Offshore composite structures

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Anode · Coating · Repaint

Nil

Grounding / Bonding
Required

25+ yr

Marine-Air Service Life

-70 %

Weight vs. Steel Racking

RELATED F1 PRODUCTS · SPEC EXCERPT

Product	Form / Size	Key Spec	Standard
FRP Support Post	SHS 50-120 mm	Corrosion-free in seawater	EN 13706
Rail & Bracing	Channel / angle	240-400 MPa; dielectric	EN 13706
Walkway Grating	Molded grating	Anti-slip; floating-PV O&M	EN ISO 14122
Custom Pultrusion	To drawing	Pontoon & fender sections	EN 13706